

Why Surface Area Isn't Just $\iint_D 1 \, dA$

In Section 15.4, we computed the area of a flat region D in the xy -plane using $\iint_D 1 \, dA$. Can we use the same formula for a **curved** surface?

No! Think of draping a bedsheet over a tent pole. The sheet covers more area than its shadow on the floor. A tilted or curved surface has **more area** than its projection onto the xy -plane.

How much more area? That depends on how **steep** the surface is. The steeper the tilt, the more surface area is packed above each little patch of the xy -plane.

The idea is the same as arc length in Calc I: a tilted line segment is longer than its horizontal projection by a factor of $\sqrt{1 + (f')^2}$. In Calc III, a tilted surface patch is bigger than its shadow by a factor of $\sqrt{f_x^2 + f_y^2 + 1}$.

Surface area = $\iint_D (\text{correction factor}) \, dA$, where the correction factor accounts for the tilt of the surface above each point.

Approximating Surface Area with Tangent Plane Patches

How do we find the correction factor? The same way we derive most integral formulas — **approximate and add up**.

Step 1: Divide the domain D into small rectangles of area $\Delta A = \Delta x \cdot \Delta y$.

Step 2: Above each rectangle, the surface $z = f(x, y)$ is approximately a **tilted parallelogram** — a piece of the tangent plane.

Step 3: The tangent plane at a point has two direction vectors:

$$\vec{a} = \langle 1, 0, f_x \rangle \Delta x \quad \vec{b} = \langle 0, 1, f_y \rangle \Delta y$$

Step 4: The area of the parallelogram is $|\vec{a} \times \vec{b}|$:

$$\vec{a} \times \vec{b} = \langle -f_x, -f_y, 1 \rangle \Delta x \Delta y$$

$$|\vec{a} \times \vec{b}| = \sqrt{f_x^2 + f_y^2 + 1} \Delta x \Delta y$$

$\sqrt{f_x^2 + f_y^2 + 1}$ is the magnitude of the **normal vector** $\langle -f_x, -f_y, 1 \rangle$ to the surface. Steeper surface $\rightarrow$ bigger normal vector $\rightarrow$ bigger correction factor.

The Surface Area Formula

Summing up the parallelogram areas and taking the limit gives us the integral formula.

Surface Area

The area of the surface $z = f(x, y)$ over a region D , where f_x and f_y are continuous, is

$$A(S) = \iint_D \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1} \, dA$$

Sanity check: If f is **constant** (a flat horizontal surface), then $f_x = 0$ and $f_y = 0$, so the integrand is $\sqrt{0 + 0 + 1} = 1$, and we get $A(S) = \iint_D 1 \, dA = \text{area of } D$. That's exactly right — a flat surface has the same area as its shadow!

The formula $\sqrt{f_x^2 + f_y^2 + 1}$ always gives a value ≥ 1 , so the surface area is always at least as large as the area of D . Equality only when the surface is perfectly flat.

Quick Check: Surface Area of a Flat Surface

You have the surface $z = 3$ (a horizontal plane) over the unit disk $x^2 + y^2 \leq 1$.

What is the surface area?

Compute f_x and f_y for $f(x, y) = 3$. Plug into the surface area formula. What does the integral simplify to?

Example 1: Plane Inside a Cylinder

Example 1

Find the surface area of the part of the plane $2x + 5y + z = 10$ that lies inside the cylinder $x^2 + y^2 = 9$.

Solve for $z = f(x, y)$, compute the partials, and set up the integral over the disk.

Example 2: Area of a Spherical Cap

Example 2

Find the surface area of the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the plane $z = 1$.

Solve for z , find f_x and f_y , simplify the integrand, then switch to polar.

Quick Check: Set Up a Surface Area Integral

Consider the paraboloid $z = x^2 + y^2$ over the unit disk $x^2 + y^2 \leq 1$.

Set up (but don't evaluate) the surface area integral in polar coordinates.

Compute f_x and f_y , form the integrand, then convert to polar.

Example 3: Why $A(\text{sphere}) = 4\pi a^2$

Example 3

Prove that the surface area of a sphere of radius a is $4\pi a^2$.

Strategy: Find the area of the upper hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and double it. But there's a catch — the integrand blows up at the equator!

Set up the integrand for $z = \sqrt{a^2 - x^2 - y^2}$ over the disk $x^2 + y^2 \leq a^2$. Where does the integrand have trouble?

Homework

Section 15.5: 1, 3, 9, 11, 13, 23